

Swiss geoawareness data provision description and testing

Document version

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The first purpose of this document is to provide details of the authoritative data provision for UAS Geographical Zones (UGZ) to USSPs from the Common Information System (CIS) in the framework of the Swiss soft launch in Zürich version 1.

Then a set of performance rules is defined for the purpose of automatically testing the capabilities of USSPs to ingest UGZs and interpreting them correctly in different given operating rules with given scenarios. For now, the objective is to test whether, in a given geospatial position at a given moment, a restriction applies.

Legal Basis

document	legal source	Legal base
Swiss Ordinance of the DETEC on special category aircraft	https://www.fedlex.admin.ch/eli/oc/2024/298/fr#mod_u7	Art. 29e, par 1
Commission Implementing Regulation (EU) 2021/664 of 22 April 2021 on a regulatory framework for the U-space	Implementing regulation - 2021/664 - EN - EUR-Lex	Art. 5, par 1
Commission Implementing Regulation (EU) 2019/947 of 24 May 2019 on the rules and procedures for the operation of unmanned aircraft	http://data.europa.eu/eli/reg_impl/2019/947/oj	Art. 15, par 3

Authoritative Data

Data sources

1. **Operating rules (OR)** require an underlying data source. The operation rules identify the authoritative data that shall be used by compliant implementation.
2. Data sources are provided through the CIS via HTTPS file transfer protocol.
3. There are two categories of data: aeronautical data and non-aeronautical data both available as **UAS Geographical Zones (UGZ)** ; the data model, structure, and format according to the swiss national implementation of the EUROCAE standard ED-318, Edit. 2024. For each category, the data type, the authoritative source, the minimum refresh rate, and the associate note are provided in the **Data sources** table.

Data sources

Data source ID	Data type	Authoritative Source	Refresh Rate (Minimum)	When applied	Notes
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[FOCA001]	5km airport zones Control zone (CTR) Reduced airport zones Perimeter Prisons Military installations (Outdoor) switchgear or substations at network level 2 of power supply facilities Ruswil compressor stations & Wallbach measuring station (i.e., natural gas supply facilities) Nuclear power plants and interim storage facility in Würenlingen Swiss National Park Water and migratory bird reserves Federal hunting ban areas Vaduz Castle Cantonal protected areas	focagis.github.io/geoawareness-cis/ed318.json	Rate: every 6 hours at 05:15, 11:15, 17:15 and 23:15 LT	Data effective date per ED-318 standard validFrom to next update validFrom+6 hours (c.f. Refresh rate)	Data model, structure and file format: EUROCAE Standard ED-318 swiss national implementation converted from EUROCAE ED-269 with Swiss implementation (see source see converter) If there are errors in the file the user is expected to try again to retrieve the data using the latest valid file update provided by FOCA in the mid time. The data type for the U-space soft launch volume will be add by oct. 2025
[SKYGUIDE001]	Control zone (CTR)	https://focagis.github.io/geoawareness-cis/skyguide_ed318.json	1h	Data effective date per ED-318 standard validFrom to next update validFrom+6 hours (c.f. Refresh rate)	Aeronautical data in ED-318 standard is available through an API combined with a DSS. The description of this data service is defined in [Skyguide SDD].

Restrictions

Restrictions ID	Description	Data Source
[REC01]	The operation of unmanned aircraft is prohibited.	[FOCA001] [SKYGUIDE001]
[REC02a]	The operation of unmanned aircraft is only allowed with exemption permit.	[FOCA001] [SKYGUIDE001]
[REC02b]	The operation of unmanned aircraft weighing more than 250 g is only allowed with exemption permit.	[FOCA001] [SKYGUIDE001]
[REC02c]	The operation of unmanned aircraft weighing more than 250 g is only permitted from an altitude of 120 m above ground with exemption permit.	[FOCA001] [SKYGUIDE001]

[REC03]	The operation of unmanned aircraft is permitted under certain conditions.	[FOCA001] [SKYGUIDE001]
[REC04]	The operation of unmanned aircraft is permitted under certain conditions.	[FOCA001] [SKYGUIDE001]
[REC05]	The operation of unmanned aircraft is permitted.	[FOCA001] [SKYGUIDE001]

Operating rules

Rule ID	Description	Note
OR010	The operation of unmanned aircraft weighing less than or equal to 250 g.	
OR011	The operation of unmanned aircraft weighing more than 250 g.	
OR020	The operation of unmanned aircraft below or at an altitude of 120 m above ground.	
OR021	The operation of unmanned aircraft above an altitude of 120 m above ground.	

Test Scenarios

For each Test scenario, a test can be associated with different expected output according to the operating rules (the identifier in square brackets refers to the applicable OR). A scenario allows a given situation to be developed that simulates an operational reality.

Test	Operating Rule	Test Scenario	Expected Output	Result	Note
1.1	[OR020]	The app identifies restricted area identifier [CTR0015] with [REC02c].	None		
	[OR021] [OR010]		None		
	[OR021] [OR011]	Location details Name : CTR ZURICH Date/Time: Anytime Lat/Long: 47.376887, 8.490805	Advise		

Test	Operating Rule	Test Scenario	Expected Output	Result	Note
1.2	[OR010]	The app identifies restricted area identifier [LSZH001] with [REC02b].	None		
	[OR011]		Advise		

		Location details Name : LSZH ZURICH Date/Time: Anytime Lat/Long: 47.411341, 8.526467			
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Test	Operating Rule	Test Scenario	Expected Output	Result	Note
1.3	All	The app identifies restricted area identifier [GZW001] with [REC01]. Location details Name : Gefängnis Zürich West Date/Time: Anytime Lat/Long: 47.381472, 8.519441	Advise		

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YAML Reference

```
- geospatial_check_id: TEST_001
  Requirement_ids: []
  description: The first test step defined by the test
  designer
    operation_rule_set: OR010
    restriction_source: FOCA001 | SKYGUIDE001
    volumes:
      - outline_circle:
          center:
            lng: 7.4774
            lat: 46.9749
          radius:
            value: 10
            units: M
          altitude_lower:
            value: 0
            units: M
            reference: SFC
          altitude_upper:
            value: 100
            units: M
            reference: SFC
          start_time:
            time_during_test: StartOfTestRun
          end_time:
            time_during_test: EndOfTestRun
        expected_result: Advise | None
```